AI-Assisted Learning and the Reversal of Long-Run Skill Degradation

William Grados*

Working manuscript for the final presentation, November 2026

Abstract

Can AI assistance improve long-run skill while displacing human effort? I preserve the myopic worker problem of Aouad, Lykouris, and Zhong (2026) and replace only the learning rate with $\lambda_0 + \kappa e(1 + \eta a)$. In a two-state continuous-time Markov chain, let $b > 0$ be low-state effort without AI. The stationary high-skill share initially increases if and only if $\eta b > 1$. Its unique maximum is then at $(b - 1/\eta)/2$, its return to the no-AI share is at $b - 1/\eta$, and effort vanishes at b. Exact pairwise comparisons distinguish marginal reversal from gains relative to no AI. Reproducible formula evaluations illustrate the mechanism; independent Markov trajectories check predicted occupation shares with explicit Monte Carlo error and initial-condition bias. Mathematical proofs are complete here. The official Lean workflow has not run; no result is reported as Lean verified.

1 Introduction

AI can substitute for human effort in current production while making the remaining effort more effective for learning. These effects need not have the same implications for the long-run distribution of skill. This paper asks when the second effect is strong enough to reverse the first. The answer is conditional and local: in the two-state model studied here, the stationary high-skill share rises over an initial interval precisely when the product of learning complementarity and low-state effort without AI exceeds one.

The starting point is Proposition 3.5 of Aouad et al. (2026, p. 8). In their endogenous-skill model, assistance reduces effort; an upward learning rate that depends increasingly on effort alone then falls. Lower assistance therefore first-order stochastically dominates higher assistance in the stationary skill distribution. I retain their worker objective, myopia, additive production inputs, and constant downward transition rate. The only modified primitive is the learning technology: AI enters it directly and complements effort. For each fixed assistance level it remains positive, increasing, and affine in effort.

The contribution is a complete two-state characterization, including the corner at zero effort. Three comparisons have different thresholds: the marginal effect of more assistance, the comparison with no assistance, and the comparison of two positive assistance levels. Confusing them would overstate what complementarity achieves. The stationary high-skill share can remain above its no-AI value after its marginal assistance effect has become negative. Sufficiently high assistance eliminates the effort channel entirely.

*Universidad del Pacífico. Track A extension of Aouad, Lykouris, and Zhong (2026). Repository: https://github.com/wbgradost/ai-project.

This is a working manuscript for critique, not a claim that the November final-paper deliverable is complete. Mathematical statements and proofs identify a future formalization target. Numerical checks and stochastic simulations respond to feedback on the topic presentation. The author's handwritten derivations and the official paper-scoped Lean run remain pending; their status is reported explicitly.

2 Relation to the baseline and established literature

The baseline is the reviewed arXiv version 1 of Aouad et al. (2026), especially Proposition 2.1 (effort), Proposition 3.5 (stationary skill degradation), and Appendix A.4 (its proof). That comparison orders stationary distributions through upward transition rates that fall with assistance. In this extension an upward rate can instead increase, even though optimal effort continues to fall. With two ordered states, the distributional comparison reduces to the high-state probability. This restriction permits an exact result; no N-state extension is asserted.

The topic-stage review distinguished this contribution from Davies (2026), a working paper in the reviewed September 2026 version 3. Davies studies course design, work versus delegation, and state-dependent effects of AI quality on second-best per-task effort and skill acquisition under his stated conditions. This paper does not claim to introduce AI–learning complementarity. It gives closed-form stationary thresholds in the different birth–death framework of Aouad et al. The experimental studies reviewed at the topic stage (Shen and Tamkin, 2026; Wu et al., 2026) motivate distinguishing active use from delegation; they do not estimate this paper's parameters or validate its stationary comparative static. Bibliographic version descriptions refer to the sources reviewed at that stage, rather than a new claim about current editorial status.

3 Model and assumptions

Assistance a is fixed and exogenous, held constant as the skill process evolves. The worker observes current skill s and chooses effort $e \geq 0$ to maximize current utility. There is no continuation value in this choice. Skill changes subsequently through the rates specified below.

Assumption 1 (Worker problem and selection)**.** The production function $p : [0, \infty) \to \mathbb{R}$ is continuously differentiable, nondecreasing, and concave. Effort costs γe, where $\gamma > 0$. The growth condition is

$$\limsup_{x \to \infty} \frac{p(x)}{x} < \gamma.$$

The worker selects the largest effort among maximizers of

$$\max_{e \geq 0} \{p(s + e + a) - \gamma e\}, \qquad s, a \geq 0. \tag{1}$$

Write $x^* = \max \arg\max_{x \geq 0} \{p(x) - \gamma x\}$.

The selection rule preserves the baseline convention under weak concavity; uniqueness is not silently imposed. The growth restriction guarantees a nonempty compact maximizer set. At an interior optimum the FOC is $p'(s + e + a) = \gamma$. Learning cannot change this condition because it is absent from the myopic objective.

Assumption 2 (States, learning, and depreciation)**.** There are two skills $0 \le s_L < s_H$, with $b := x^* - s_L > 0$. For fixed $a \ge 0$, the skill process is a time-homogeneous continuous-time Markov chain. Its transition rate from L to H is

$$\lambda(e, a) = \lambda_0 + \kappa e(1 + \eta a), \qquad \lambda_0 > 0, \quad \kappa > 0, \quad \eta \ge 0, \tag{2}$$

evaluated at optimal low-state effort. The rate from H to L is the constant $\mu > 0$. There is no state above H.

High-state effort can affect current production, but cannot generate another upward transition in this chain. Baseline learning λ_0 does not require measured effort. The parameter κ converts effective effort into a transition intensity; η is the increase in the learning effectiveness of effort per unit of assistance. At fixed a, $\lambda_e = \kappa(1 + \eta a) > 0$ and $\lambda_{ee} = 0$; $\lambda_{ea} = \kappa\eta \ge 0$. Joint concavity in (e, a) is neither assumed nor used.

Definition 1 (Effective effort)**.** For $a \ge 0$, define

$$f(a) = (b - a)^+(1 + \eta a), \qquad \lambda_L(a) = \lambda_0 + \kappa f(a). \tag{3}$$

Let $\pi_H(a)$ be the stationary probability of H, and let $\pi_L(a) = 1 - \pi_H(a)$.

Lemma 1 (Effort under the baseline selection rule)**.** *Under Assumption 1, selected effort is*

$$e^*(s, a) = (x^* - s - a)^+. \tag{4}$$

Under Assumption 2, $e_L^*(a) = b - a > 0$ for $0 \le a < b$; $e_L^*(a) = 0$ for $a \ge b$. On $(0, b)$ its derivative is -1.

Proof. Set $z = s + a$ and $q(x) = p(x) - \gamma x$. Replacing e with $x = z + e$ rewrites the objective as $q(x) + \gamma z$ on $x \ge z$. The growth restriction gives $q(x) \to -\infty$. Continuity gives a nonempty compact global maximizer set, with largest member x^*. If $z \le x^*$, this member is feasible and remains the largest constrained maximizer. If $z > x^*$, q is strictly decreasing on $[z, \infty)$. Indeed, take $x^* < x < y$. If $q(y) \ge q(x)$, concavity would imply

$$\frac{q(x) - q(x^*)}{x - x^*} \ge \frac{q(y) - q(x)}{y - x} \ge 0,$$

contradicting $q(x) < q(x^*)$, since x is beyond the largest global maximizer. The constrained maximizer is then z. Thus the selected total input is $\max\{x^*, z\}$, yielding (4). Substitute $s = s_L$ and use $b > 0$ for the last claims. Since $x^* > s_L \ge 0$, differentiability at the interior maximizer also gives $p'(x^*) = \gamma$. No strict concavity is needed. □

Lemma 2 (Stationarity and finite-horizon occupation)**.** *Under Assumptions 1–2, each fixed $a \ge 0$ has a unique stationary distribution,*

$$\pi_H(a) = \frac{\lambda_L(a)}{\lambda_L(a) + \mu}, \qquad \pi_L(a) = \frac{\mu}{\lambda_L(a) + \mu}. \tag{5}$$

Write $q_a = \lambda_L(a) + \mu > 0$. If $h_0 \in [0, 1]$ is the initial probability of H, its probability at time $t \ge 0$ is

$$h(t) = \pi_H(a) + (h_0 - \pi_H(a))e^{-q_a t}. \tag{6}$$

For $T > 0$, the occupation fraction $Y_T = T^{-1} \int_0^T \mathbf{1}_{\{X_t = H\}}\, dt$ satisfies

$$\mathbb{E}[Y_T] = \pi_H(a) + (h_0 - \pi_H(a))\frac{1 - e^{-q_a T}}{q_a T}. \tag{7}$$

Starting in L therefore creates a downward bias of $\pi_H(a)(1 - e^{-q_a T})/(q_a T)$ relative to stationarity.

Proof. The generator in order (L, H) is

$$Q(a) = \begin{pmatrix} -\lambda_L(a) & \lambda_L(a) \\ \mu & -\mu \end{pmatrix}.$$

Both off-diagonal rates are strictly positive by Assumption 2; the finite chain is irreducible. Stationary balance and normalization are $\pi_L \lambda_L = \pi_H \mu$ and $\pi_L + \pi_H = 1$. They imply $(\lambda_L + \mu)\pi_H = \lambda_L$. Division is valid because $\lambda_L \geq \lambda_0 > 0$ and $\mu > 0$. The resulting probabilities are positive, sum to one, and uniquely solve the two equations. The forward equation is $h' = \lambda_L(1-h) - \mu h = \lambda_L - q_a h$. Subtract $\pi_H = \lambda_L / q_a$ to obtain $(h - \pi_H)' = -q_a(h - \pi_H)$, with unique solution (6). Boundedness of the indicator allows expectation and integration to be interchanged. Integrating (6) on $[0, T]$ and dividing by T proves (7). Set $h_0 = 0$ for the stated bias. These formulas show convergence of the marginal probability and the expected occupation to stationarity. □

4 Results: three distinct comparisons

For $\mu > 0$ the map $g(\lambda) = \lambda/(\lambda + \mu)$ is strictly increasing, with $g'(\lambda) = \mu/(\lambda + \mu)^2 > 0$. Every stationary comparison therefore has the sign of its learning-rate comparison. Positivity of μ matters: with $\mu = 0$ irreducibility is lost and the stationary share no longer encodes these strict comparisons.

Proposition 1 (Global shape, maximum, and comparison with no AI)**.** *Under Assumptions 1–2, π_H is continuous on $[0, \infty)$. For $0 \leq a < b$, its derivative (the right derivative at 0) is*

$$\pi'_H(a) = \frac{\mu\kappa[\eta(b - 2a) - 1]}{[\lambda_L(a) + \mu]^2}. \tag{8}$$

The following statements hold.

1. *If $\eta b \leq 1$, π_H is strictly decreasing on $[0, b]$, has its unique global maximum at $a = 0$, and $\pi_H(a) < \pi_H(0)$ for all $a > 0$. If $\eta b = 1$, $\pi'_H(0+) = 0$ but $\pi'_H(a) < 0$ on $(0, b)$.*

2. *If $\eta b > 1$, define*

$$a^* = \frac{b - 1/\eta}{2}, \qquad a^R = b - 1/\eta = 2a^*. \tag{9}$$

Then $0 < a^* < a^R < b$ and $a^* < b/2$; π_H is strictly increasing on $[0, a^*]$, strictly decreasing on $[a^*, b]$, and has its unique global maximum at a^*. Moreover,

$$\pi_H(a) > \pi_H(0) \iff 0 < a < a^R, \qquad \pi_H(a^R) = \pi_H(0), \qquad \pi_H(a) < \pi_H(0) \quad (a > a^R). \tag{10}$$

3. *In either regime, for all $a \geq b$, effort is zero and*

$$\lambda_L(a) = \lambda_0, \qquad \pi_H(a) = \frac{\lambda_0}{\lambda_0 + \mu} < \pi_H(0). \tag{11}$$

At b the one-sided derivatives are

$$\pi'_H(b-) = -\frac{\mu\kappa(1 + \eta b)}{(\lambda_0 + \mu)^2} < 0, \qquad \pi'_H(b+) = 0. \tag{12}$$

On $(0, b)$, $\pi''_H < 0$; this does not imply global concavity across b.

Proof. On $[0, b)$ expand

$$f(a) = b + (\eta b - 1)a - \eta a^2, \qquad f'(a) = \eta(b - 2a) - 1, \qquad f''(a) = -2\eta. \tag{13}$$

The quotient and chain rules give (8); its denominator is positive and $\mu\kappa > 0$. When $\eta b \leq 1$, $f'(a) < 0$ for every $a \in (0, b)$: if $\eta = 0$, it equals -1, and otherwise $\eta b - 1 - 2\eta a < 0$. The mean value theorem gives strict decrease, including endpoint 0 even when its right derivative vanishes. At b, $f(a) > 0 = f(b)$ for each $a < b$ extends the strict comparison.

When $\eta b > 1$, necessarily $\eta > 0$. Solving $f'(a) = 0$ gives a^*. The condition $b > 1/\eta$ gives $a^* > 0$; $1/\eta > 0$ gives $a^* < b/2$, and $0 < a^* < 2a^* = a^R < b$. The derivative is positive before a^* and negative after it. Strict monotonicity of g transfers these comparisons to π_H. For $0 < a < b$, subtract $f(0) = b$:

$$f(a) - b = a[\eta(b - a) - 1]. \tag{14}$$

If $\eta b \leq 1$, the bracket is negative for $a > 0$. If $\eta b > 1$, it is positive exactly for $a < a^R$, zero at a^R, and negative thereafter. This proves the active-region no-AI comparisons. At $a = 0$ the shares agree, so a strict gain excludes zero.

For $a \geq b$, the positive part in (3) vanishes, proving (11). Since $f(0) = b > 0$, the plateau lies strictly below the no-AI share. The positive part is continuous, and the rational denominator remains positive, proving continuity at b. Take $a \uparrow b$ in (8) to obtain the left derivative in (12). The right derivative is zero because π_H is constant on $[b, \infty)$. These signs and the plateau also establish uniqueness of the claimed global maxima. On $(0, b)$,

$$\pi_H''(a) = -\frac{2\mu\kappa\eta}{(\lambda_L + \mu)^2} - \frac{2\mu\kappa^2[f'(a)]^2}{(\lambda_L + \mu)^3} < 0. \tag{15}$$

If $\eta > 0$, the first term is strictly negative; if $\eta = 0$, $f' = -1$ makes the second strictly negative. The upward derivative jump at b prevents a global concavity conclusion. □

Proposition 2 (Comparison of two assistance levels)**.** *Under Assumptions 1–2, for any $0 \leq u < v$, with $D_w = \lambda_0 + \kappa f(w) + \mu > 0$,*

$$\pi_H(v) - \pi_H(u) = \frac{\mu\kappa[f(v) - f(u)]}{D_v D_u}. \tag{16}$$

If $v \leq b$, then

$$\pi_H(v) - \pi_H(u) = \frac{\mu\kappa(v - u)[\eta(b - u - v) - 1]}{D_v D_u}. \tag{17}$$

Thus $\pi_H(v) > \pi_H(u)$ if and only if $\eta(b - u - v) > 1$; the analogous equality and strict-reverse equivalences also hold. If $u < b \leq v$, $\pi_H(v) < \pi_H(u)$. If $b \leq u < v$, the shares agree. In two states, the share comparison also characterizes first-order stochastic dominance and the ordering of mean skill.

Proof. Subtract the fractions in (5). The common numerator is $\lambda_L(v)[\lambda_L(u)+\mu]-\lambda_L(u)[\lambda_L(v)+\mu] = \mu[\lambda_L(v) - \lambda_L(u)]$, proving (16). For $v \leq b$, the quadratic in (13) is valid at both points, including $v = b$, so

$$f(v) - f(u) = (\eta b - 1)(v - u) - \eta(v^2 - u^2) = (v - u)[\eta(b - u - v) - 1].$$

The factors $D_u D_v$, $\mu\kappa$, and $v - u$ are positive, yielding all sign equivalences. If $u < b \leq v$, then $f(u) > 0 = f(v)$; if $b \leq u < v$, both equal zero. Finally, the cumulative probability between s_L and s_H is $1 - \pi_H$, and expected skill is $s_L + (s_H - s_L)\pi_H$. Since $s_H > s_L$, these comparisons are exactly the share comparison. This is a two-state statement. □

An increase from u to v can improve skill even when v exceeds the maximum: in the strong regime the exact condition is $u+v<2a^*$. A marginal comparison at v instead requires $v<a^*$. The no-AI comparison sets $u=0$ and requires $v<2a^*$. These statements answer different questions; none says that every positive increase improves skill.

Corollary 1 (Threshold locations and outcome magnitudes)**.** *Under Assumptions 1–2, critical complementarity is $\eta_c=1/b$. In the strong regime the thresholds a^* and a^R depend only on η, b, independently of λ_0, κ, μ. Holding the other arguments fixed,*

$$\partial_\eta a^* = \frac{1}{2\eta^2}, \quad \partial_\eta a^R = \frac{1}{\eta^2}, \quad \partial_b a^* = \frac{1}{2}, \quad \partial_b a^R = 1. \tag{18}$$

For fixed $a \geq 0$ and $D=\lambda_L(a)+\mu$,

$$\partial_{\lambda_0}\pi_H = \frac{\mu}{D^2} > 0, \qquad \partial_\kappa \pi_H = \frac{\mu f(a)}{D^2} \geq 0, \qquad \partial_\mu \pi_H = -\frac{\lambda_L(a)}{D^2} < 0. \tag{19}$$

Also $\partial_\eta \pi_H = \mu\kappa a(b-a)^+/D^2 \geq 0$ (a right derivative at $\eta=0$); for $a<b$, $\partial_b \pi_H = \mu\kappa(1+\eta a)/D^2 > 0$. Multiplying all three rates (λ_0, κ, μ) by the same $c>0$ leaves the entire stationary curve unchanged and multiplies the convergence rate q_a by c.

For a fixed $0<a<a^R$ in the strong regime, the positive no-AI gain is

$$G(a) := \pi_H(a) - \pi_H(0) = \frac{\mu\kappa\Delta}{(\lambda_0+\kappa f(a)+\mu)(\lambda_0+\kappa b+\mu)}, \qquad \Delta = f(a)-b>0. \tag{20}$$

It strictly decreases with λ_0. Its derivative in μ has the sign of $(\lambda_0+\kappa f(a))(\lambda_0+\kappa b)-\mu^2$; its derivative in κ has the sign of $(\lambda_0+\mu)^2-\kappa^2 f(a)b$. Thus stronger effort sensitivity or faster depreciation need not monotonically increase the gain, even though they leave its assistance thresholds unchanged.

Proof. The regime boundary and thresholds follow from Proposition 1; differentiation gives (18). Their independence of the other rates is literal. Apply the quotient rule in (5), using $\partial_\kappa \lambda_L = f$, $\partial_{\lambda_0}\lambda_L = 1$, $\partial_\eta \lambda_L = \kappa a(b-a)^+$, and, for $a<b$, $\partial_b \lambda_L = \kappa(1+\eta a)$, to obtain the level formulas. For common scaling, cancel c in $c\lambda_L/(c\lambda_L+c\mu)$; the sum of the rates becomes cq_a, proving the convergence statement using Lemma 2. Setting $u=0$ in (16) proves (20). Let $C_1=\lambda_0+\kappa f(a)>0$, $C_0=\lambda_0+\kappa b>0$, $M=(C_1+\mu)(C_0+\mu)>0$, and $B=\lambda_0+\mu>0$. Holding a, b, η fixed, differentiation gives

$$\partial_{\lambda_0} G = -\frac{\mu\kappa\Delta[(C_1+\mu)+(C_0+\mu)]}{M^2} < 0, \qquad \partial_\mu G = \frac{\kappa\Delta(C_1C_0-\mu^2)}{M^2},$$

$$\partial_\kappa G = \frac{\mu\Delta[B^2-\kappa^2 f(a)b]}{M^2}.$$

All prefactors and denominators are positive. This establishes the stated signs and equality cases without assuming a globally monotone gain. □

Threshold invariance holds conditional on fixed b, η, not on arbitrary changes to the worker problem. For the numerical example $p(x)=A\log(1+x)$, $A>\gamma>0$, one has $x^*=A/\gamma-1$ and $b=A/\gamma-1-s_L>0$. Raising A increases b; raising γ or s_L lowers it. The stationary curve depends on the dimensionless arguments a/b, ηb, λ_0/μ, and $\kappa b/\mu$, as follows by dividing (5) by μ and factoring b from f.

5 Numerical evidence and stochastic validation

5.1 Design and economic interpretation

All parameters are illustrative, not estimated or calibrated to data. Use $p(x) = 4\log(1+x)$, $\gamma = 1$, $s_L = 1$, and $s_H = 3$; then $x^* = 3$ and $b = 2$. Set $\lambda_0 = 0.1$, $\kappa = 0.4$, and $\mu = 0.3$, with rates per chosen time unit. Without measured effort the mean upward waiting time is ten units; the mean high-state spell is $1/0.3$ units. These units have no assigned calendar interpretation. Assistance and effort have the same production-input units.

We compare $\eta \in \{0, 0.25, 1\}$, so $\eta b \in \{0, 0.5, 2\}$. The first regime excludes complementarity; the second allows it below the critical value; the third crosses the condition in Proposition 1. A separate equality check uses $\eta = 0.5$, testing the zero derivative at $a = 0$ without misclassifying it as a gain region.

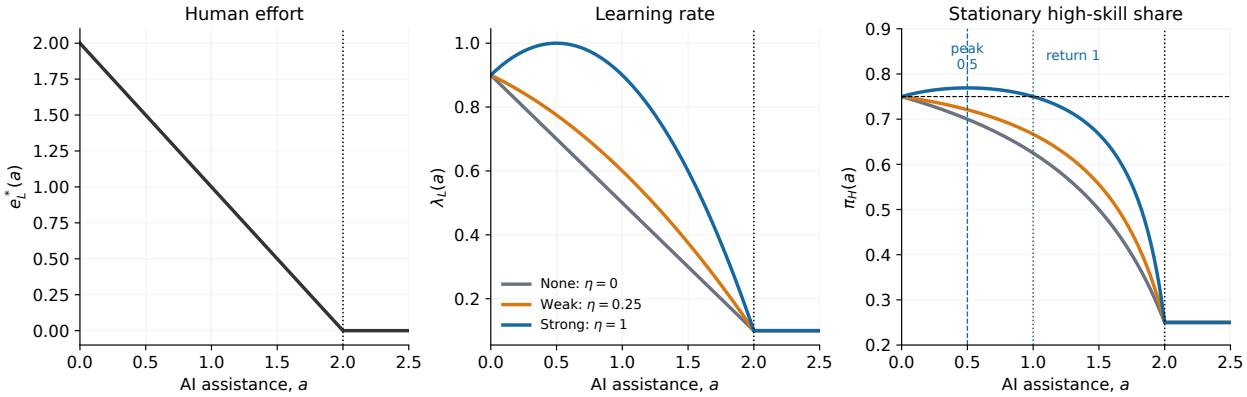


Figure 1: Formula evaluations of effort, learning, and stationary skill. Effort is identical across regimes. Only strong complementarity makes the learning rate and high-skill share rise initially. These curves are not stochastic simulations. Source: `code/verify.py`.

Figure 1 separates behavior from learning. Effort falls one-for-one until $a = 2$ in every regime: the learning technology does not alter that response. Without complementarity learning falls immediately. Weak complementarity attenuates the fall but cannot reverse it. Strong complementarity raises learning for $0 < a < 0.5$ even as effort falls. Beyond the maximum, additional effectiveness no longer offsets displaced effort. Every regime meets the same zero-effort plateau.

In the strong example, the maximum is at $a = 0.5$, the return to the no-AI share at $a = 1$, and effort disappears at $a = 2$. The no-AI share is $3/4$; the maximum is $10/13$, a gain of $1/52$, approximately 1.9231 percentage points. The plateau is $1/4$. Indeed, $f(0) = 2$, $f(0.5) = 2.25$, $\lambda_L(0) = 0.9$, and $\lambda_L(0.5) = 1.0$. Between 0.5 and 1, assistance lowers skill at the margin while the share remains above its no-AI value.

5.2 Thresholds and sensitivity

Figure 2 shows which parameters determine locations. A larger ηb creates and expands the gain regions, but the marginal maximum always lies below $b/2$. The no-AI return threshold approaches b as complementarity grows, but never reaches it for finite η. The rates λ_0, κ, μ do not move these boundaries: they multiply or monotonically transform the same effective-effort comparison. Production parameters can move them through b. Figure 3 separates location from magnitude. Baseline learning raises all shares; depreciation lowers all shares. Neither changes where $\pi_H'(a)$

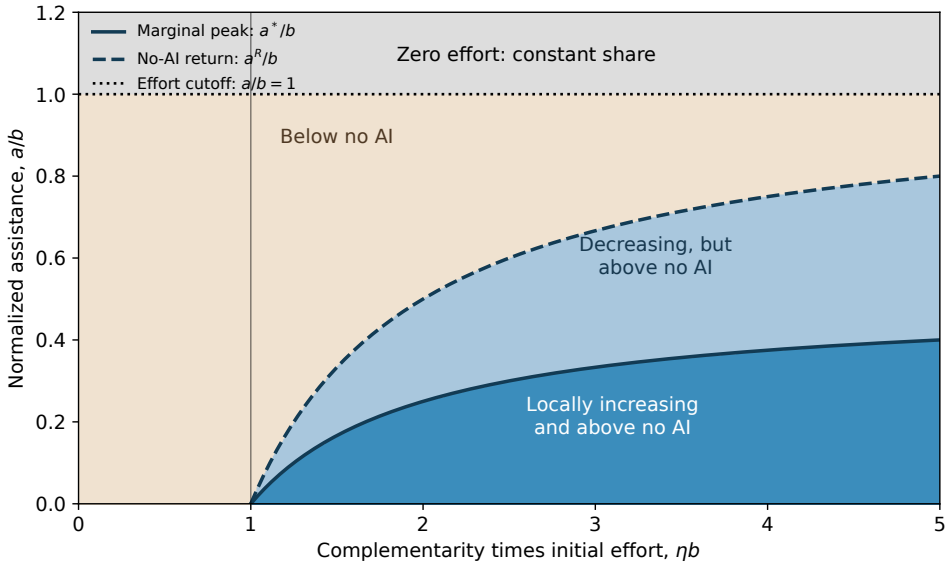


Figure 2: Normalized threshold map. The dimensionless coordinates ηb and a/b locate marginal improvement, improvement over no AI, and zero effort. The two improvement thresholds are distinct. This is a formula evaluation, not Monte Carlo evidence. Source: `code/verify.py`.

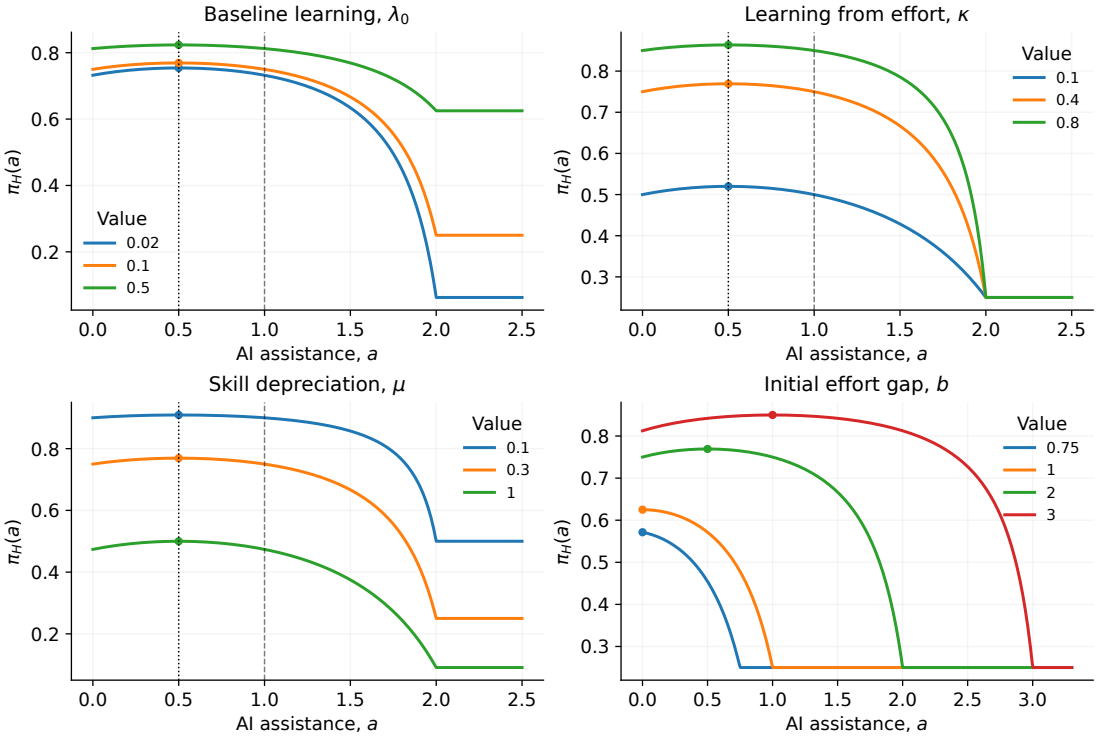


Figure 3: Sensitivity of the stationary outcome to rate parameters and the effort primitive. Rate changes alter levels and gains while preserving assistance thresholds. Changes in b alter the reversal condition and threshold locations. Source: `code/verify.py`.

changes sign. A large stationary share need not imply a large complementarity gain: high baseline learning compresses the gain over no AI. The gain can also be small when increased effort sensitivity drives both compared shares toward one. Corollary 1 explains why the gain need not be monotone in κ or μ.

5.3 Continuous-time Markov simulations

Each path draws an exponential holding time with its current state's exit rate, then switches to the other state. There is no discrete-time transition approximation. Occupation is time spent in H, including the final censored holding spell, divided by the horizon. Counting the proportion of visits to H would be inappropriate: the embedded jump chain alternates and does not have the same occupation law.

For each of 21 assistance/regime cells, independent paths start in L. There are 384 replications per cell and horizons $T = 25, 100, 400$. The cells combine the three complementarity values with $a \in \{0, 0.25, 0.5, 1, 1.5, 2, 2.5\}$. The master seed is 20261104; independent child streams make the experiment reproducible. No burn-in is hidden. Each path is evaluated at all three horizons, so estimates at different horizons share a trajectory; errors at a given horizon are computed across independent paths. Means are compared with the stationary share and with the exact finite-horizon expectation from Lemma 2. The standard error is the sample standard deviation of independent path occupation fractions divided by the square root of the replication count. Plotted 95% intervals use this replication-level error, rather than treating autocorrelated within-path observations as independent.

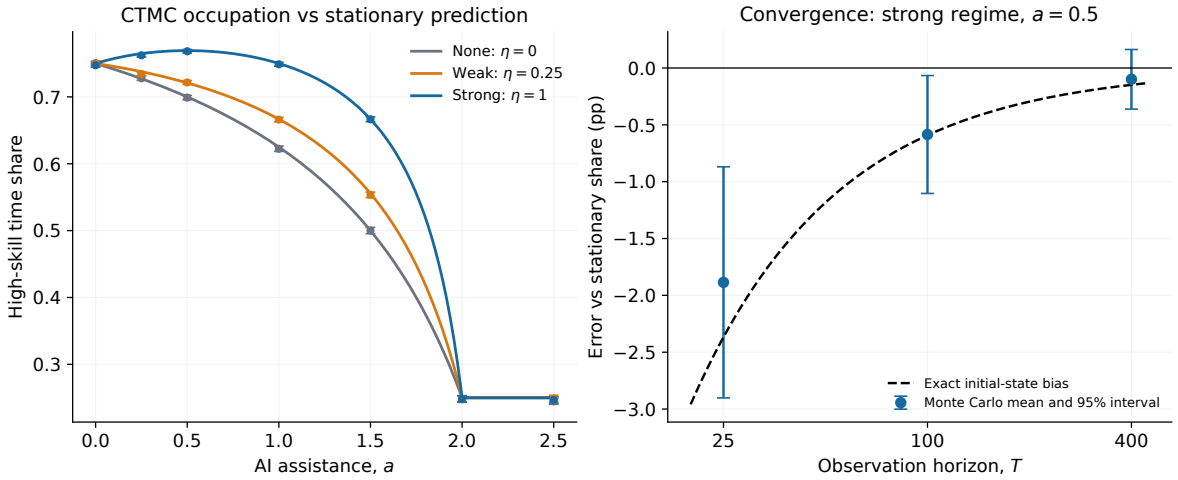


Figure 4: Stochastic occupation estimates and analytical predictions. Points and Monte Carlo intervals use independent continuous-time paths. The finite-horizon mean accounts for starting in L; stationary values are limiting predictions. Their gap can reflect both Monte Carlo error and initial-condition bias. Source: `code/verify.py`.

Figure 4 checks whether simulated dynamics reproduce the analytical occupation target, and whether increasing the horizon reduces the known initial-condition bias. At $T = 400$, the maximum absolute gap from stationarity across the simulated cells is 0.5223 percentage points, and the largest gap from the finite-horizon mean is 0.3661 percentage points. The largest standardized finite-horizon error across the checked horizons is 1.8486 exact standard errors. These are realized implementation diagnostics, not an estimate of an economic effect. Exact formula evaluation illus-

trates a proven comparative static; the stochastic experiment checks the predicted law of the chain. Neither substitutes for the proof, and finitely many paths cannot by themselves establish a long-run limit.

5.4 Checks, reproducibility, and limits

The single entry point is

```
python code/verify.py
```

Required packages are listed in `code/requirements.txt`. The script checks symbolic derivatives and difference factorizations, optimal effort against independent numerical maximization, balance and normalization, threshold and boundary cases, pairwise orderings, and parameter sensitivity. It fails on a violated claim, rather than merely saving figures. Generated tables, numeric results, and figures are stored in `code/output/` and `code/figures/`. Committed generated assets are also available to the course's independent LaTeX workflow; its configuration is unchanged.

Monte Carlo means are checked within six exact standard errors of the finite-horizon target. Exact variances follow from the integrated joint probability of being in H at two times. The diagnostic does not demand that all nominal 95% intervals contain their targets: some noncoverage is expected across multiple cells under a correct implementation. Those intervals are normal approximations across bounded independent replications; the wider failure tolerance is a reproducible diagnostic. Numerical tolerances, seeds, per-cell errors, and comparisons are recorded in the code. These experiments are illustrations, not inference about estimated economic parameters.

6 Discussion and limitations

The mechanism is a contest between two terms:

$$\lambda_L'(a) = \underbrace{\kappa\eta(b-a)}_{\text{effectiveness of retained effort}} - \underbrace{\kappa(1+\eta a)}_{\text{displaced effort}}, \qquad a < b.$$

The FOC never changes. Complementarity affects the return generated by the effort the myopic worker still chooses. The condition $\eta b > 1$ thus depends on both the technology and no-AI effort. It is not a welfare theorem, an empirical estimate, or a global recommendation for assistance. Output, effort costs, stationary skill, and social welfare need not share an ordering. The maintained interiority condition $b > 0$ is essential for a strict reversal: if $b \leq 0$, the same effort rule gives $e_L^*(a) = 0$ for every $a \geq 0$, and the stationary share is constant. Positive κ and μ are also needed for strict comparative statics; the proof does not cover a model without depreciation or without an effort-sensitive learning channel.

The two-state restriction is substantial. The value of s_H does not enter transition rates or the high-state share, although it affects mean skill and production. With more states, different effort cutoffs could make upward rates move in opposite directions; a single high-state probability would not characterize the stationary ordering. A third-state illustration is omitted to keep the verified contribution complete and interpretable.

The proof resolves stationarity, derivative signs, equality cases, the effort boundary, and all pairwise comparisons. Delicate steps were preserving the weak-concavity selection rule and distinguishing the maximum from the no-AI return. Extending the active quadratic beyond b would produce negative effort and a false global result. Global concavity is also false because the derivative jumps upward to zero at b. Operationally pending are independent author-written verification, the

official Lean run against a fixed PDF, and incorporation of class critique. No unexecuted Lean fragment is offered as a certificate.

7 Conclusion

AI–effort complementarity reverses stationary skill degradation over an initial assistance interval exactly when $\eta b > 1$. The marginal maximum, return to the no-AI benchmark, and zero-effort boundary are different objects. Analytical proofs establish these comparisons; formula evaluations explain their magnitudes, while continuous-time simulations test occupation behavior with explicit Monte Carlo error and finite-horizon bias. This manuscript defines substantive targets for the course's subsequent paper-scoped Lean procedure.

References

Ali Aouad, Thodoris Lykouris, and Huiying Zhong. Human–AI productivity paradoxes: Modeling the interplay of skill, effort, and AI assistance, 2026. arXiv preprint, version 1; conference version at the 27th ACM Conference on Economics and Computation (EC 2026).

Benjamin Davies. Course design in the age of AI, 2026. Working paper, version 3, September 2026.

Judy Hanwen Shen and Alex Tamkin. How AI impacts skill formation, 2026. Preprint, version 2.

Shang Wu, Catarina G. Belem, Shuyuan Fu, Mark Steyvers, and Padhraic Smyth. How AI assistance affects human skill development: A study of learning with logic puzzles, 2026. Preprint; accepted at Human–AI Complementarity and Alignment (HCOMP) 2026.

A Formalization targets and truthful status

No paper commit has yet been selected for formalization. No corresponding Lean declaration, generated proof file, or paper-scoped check output exists for this manuscript. Each row below is *open*: the operational blocker is that the official workflow has not started. Mathematical proofs are complete in the body; this table concerns formal verification status.

Paper result	Substantive formalization target	Declaration / file / status
Lemma 1	Largest-maximizer effort rule, weak concavity, binding constraint	None / none / open
Lemma 2	Unique stationary law; transient and occupation expectations	None / none / open
Proposition 1	Strict monotonicity, unique maximum, no-AI comparison, boundary kink	None / none / open
Proposition 2	Pairwise sign equivalence, saturation cases, two-state dominance	None / none / open
Corollary 1	Threshold invariance; level and gain derivatives	None / none / open

After author review, merge the chosen `paper/paper.pdf` into `main` and record its full commit hash. Update the existing AppliedModelingLib clone. At its root, invoke the `paper-formalization` skill on the commit-pinned PDF. Use the course-specified agent configuration. The proposed paper folder is `Grados26AIAssistedLearning`. Run

```
python3 scripts/paper_contribution.py check Grados26AIAssistedLearning -fast
```

and retain the actual output. Copy the entire generated `papers/Grados26AIAssistedLearning/` folder into `lean/`, respecting its ignore rules without forced staging. Fill the table with actual declarations, files, the pinned commit, and the status of each result: proved, proved under a stated added hypothesis, or open with the exact blocker. Re-run the workflow if a claim changes. The current PR remains unmerged for review.

Checking only fraction subtraction would be insufficient. A faithful formalization must include the positive-part boundary, positive denominators, strict-order and equality cases, and the maintained assumptions. Any added smoothness or concavity must be disclosed instead of silently presented as the exact paper statement. Assuming the desired ordering as a hypothesis would not verify the economic result. For Lemma 2, proving the displayed differential equation and its integral alone certifies formulas conditional on that equation. To claim the full Markov-chain result, the formalization must also connect the generator to the transition law and the occupation expectation. If that connection remains assumed rather than proved, the correspondence table must state the added assumption or the remaining open part.

B Critical AI check and handwritten appendix

The priority author check is

$$\pi_H(v) - \pi_H(u) = \frac{\mu\kappa(v-u)[\eta(b-u-v)-1]}{D_v D_u}, \qquad 0 \le u < v \le b.$$

It should be derived by hand from stationary balance, displaying fraction subtraction, cancellation, difference of squares, denominator positivity, and sign division. Setting $u = 0$ should give $a^R = 2a^*$, followed by a separate check for $a \ge b$. This tests the consequential distinction between local reversal and a gain relative to no AI. No handwritten evidence has been supplied; no handwritten verification verdict is claimed. The typed proof is an agent-assisted mathematical derivation, not a substitute for the required appendix in `hand/`.

AI-assisted claim	Current verdict	Evidence and remaining check
Active derivative and pairwise condition from the topic stage	Confirmed by rederivation	Complete proofs here and symbolic/numerical checks; author hand check pending.
An initial planning value of the strong-case peak share was 0.785714	Incorrect; corrected before delivery	Direct substitution gives $\lambda_L(0.5) = 1$ and $\pi_H(0.5) = 10/13 = 0.769230\ldots$.
The propositions are Lean verified	Not claimed; not verified	No official run, declarations, or check output exists; author approval of a paper version comes first.

Actual prompts, relevant responses, and corrections are recorded in `prompts.md`; no earlier conversation or author intervention is reconstructed. Numerical checks do not become a human-verification verdict.